TEK306- Assignment–Week42

(1) In the context of the lecture in Week42, explain what powerful Android applications are?

Some applications on a device have a much higher degree of power over the OS than others. This power could come through the permissions granted to them. A good example of a powerful permission that can only be granted to applications is INSTALL_PACKAGES.

(2) Mention three types of client-side exploitable software installed a host which are potentially vulnerable to remote attack.

Browsers, document readers, and email clients are vulnerable to this type of attack.

(3) Explain the principle of least exposure.

By least exposure it means that the fewer entry points there are into an application, the smaller the attack surface is. Thus, to minimize the attack surface, the developer needs to perform the activities below.

- Find every single portion of the application code that is exposed in some way to input from outside sources.
- Remove any entry points that can be.
- If an entry point has to be exposed, perform security checks at the entry points before running other code.

(4) What are the four components whose functions need to be traced for potential vulnerabilities?

The components are: Activity, Broadcast receiver, Content provider, and Service.

(5) In the lecture in Week42, we mentioned that performing security checks will only slow down a skilled reverse engineer from analyzing an app. Explain why?

The main reason is that any compiled Android application can easily be decompiled into readable source code, anyway. So, all security measures should aim to add more efforts to the one who performs reverse engineering.